

Chapter 5: Results And Evaluation

5.1 Presentation of Findings

The evaluation process for the AI-Powered Medical Imaging Assistant involved a comprehensive multistage evaluation process. The system was assessed on three dimensions: Image Quality Enhancement (Preprocessing), Diagnostic Accuracy (Classification), and Operational Efficiency/Latency.

5.1.1 Preprocessing Pipeline Evaluation

1. U-Net Enhancement Results

The U-Net-based enhancement stage has been evaluated using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) on the validation dataset as performance metrics.

Results:

```
===== U-Net ENHANCEMENT TEST RESULTS =====  
Average PSNR: 32.453 dB  
Average SSIM: 0.9066
```

Figure 5.1: U-NET Model Average PSNR and SSIM

Interpretation: With a PSNR score of 32.453 dB, the U-Net-based model provides an extremely high-quality structural enhancement due to the significant amount of noise reduction and increase in the overall clarity of each image. The model was able to maintain the structure of anatomical anatomy within the image (i.e., the outline of a malignant tumour and soft tissue), as evidenced by the SSIM metric score of 0.9066, indicating that the U-Net is a reliable pre-processing method for later classification, super-resolution, and denoising processes.

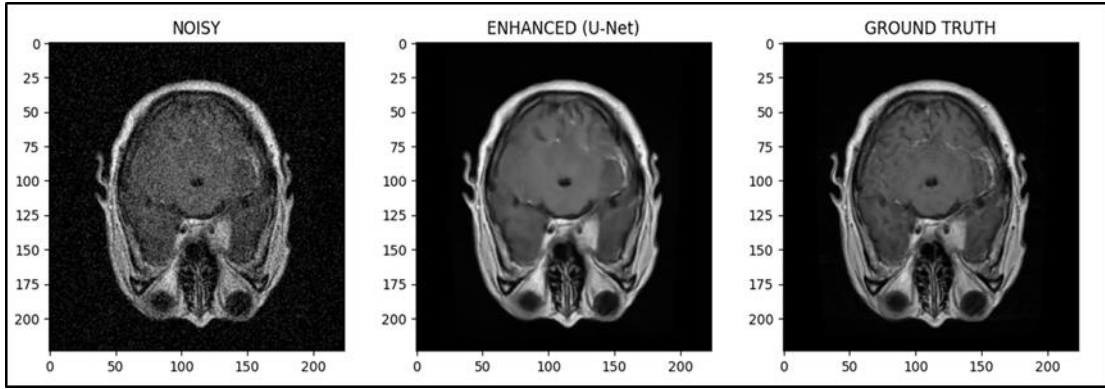


Figure 5.2: U-net Model Comparison of Noisy, Enhanced, and Ground Truth Images

2. Results of DnCNN Denoising

By applying DnCNN to the raw MRI scans to remove Rician noise artifacts, we were able to objectively assess our results against ground-truth images.

Results:

```
===== DnCNN ENHANCEMENT TEST RESULTS =====
Average PSNR: 31.701 dB
Average SSIM: 0.8919
```

Figure 5.3: Dn-CNN Model Average PSNR and SSIM

Interpretation: The Structural Similarity Index (SSIM) of 0.8919 confirms that DnCNN was able to preserve important structural features during the denoising process, whilst its Peak Signal to Noise Ratio (PSNR) of 31.701 dB shows how effectively DnCNN was able to remove noise from these images. Ultimately, the combination of smoothing and detail retention allows for these improved MRI images to remain usable when performing accurate super-resolution and diagnostic assessments.

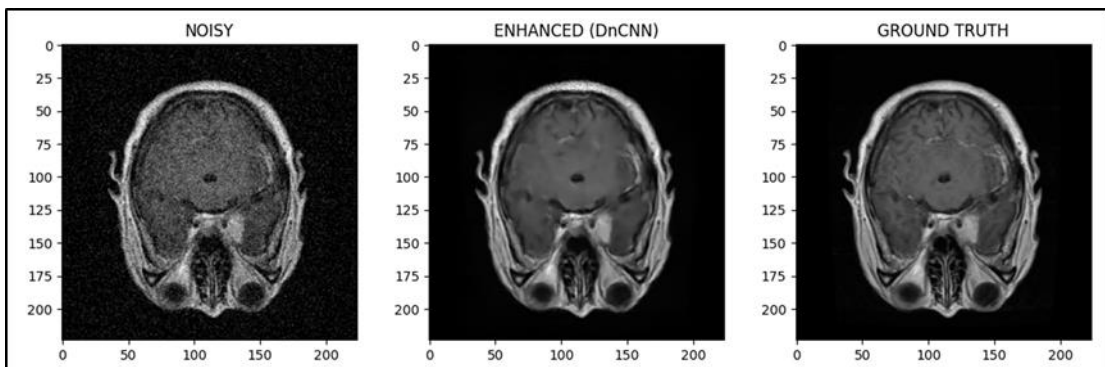


Figure 5.4: Dn-CNN Model Comparison of Noisy, Enhanced, and Ground Truth Images

3. Super-Resolution EDSR Outcomes

To upscale MRI images after denoising and reconstructing more of the high-frequency structure detail, the Enhanced Deep Super Resolution Network (EDSR) was used.

Results:

```
===== EDSR ENHANCEMENT TEST RESULTS =====  
Average PSNR: 32.469 dB  
Average SSIM: 0.8978
```

Figure 5.5: EDSR Model Average PSNR and SSIM

Interpretation: The EDSR Achieved a peak signal-to-noise ratio (PSNR) of 32.469 dB, providing a higher-quality reconstructed image than previous ANNs. The EDSR produced images that exhibited high levels of consistency with anatomical structures (i.e., without the typical "hallucination" artefacts produced by GAN-based techniques), indicated by the structural similarity index measurement (SSIM) score of 0.8978. As such, the EDSR is validated as a strong, reliable super-resolution module for this enhancement pipeline.

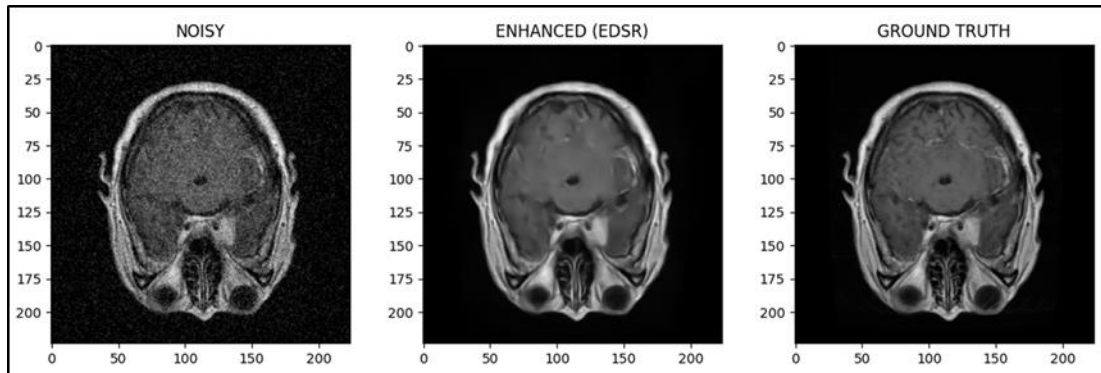


Figure 5.6: EDSR Model Comparison of Noisy, Enhanced, and Ground Truth Images

Comparing Models based on SSIM and PSNR

To assess how well the three deep-learning models (U-Net, DnCNN, and EDSR) perform in enhancing MRI images, a quantitative analysis using the PSNR and SSIM metrics was performed. The outcomes from this analysis are presented using bar charts.

1. Comparison of PSNR

When comparing the PSNR metric, the PSNR for each of the enhanced/reconstructed MRI images that were produced by the three models relative to the MRI reference images were calculated. Higher PSNR values indicate that the MRI images produced via the deep-learning

models had improved noise reduction and improved distortion characteristics compared to the MRI reference images.

Graph observations: To compare the PSNR of each of the three models, we found that U-Net had a PSNR value of ~ 32.5 dB, which is an excellent enhancement performance and also demonstrates the ability to maintain the fine anatomical details of the enhanced/reconstructed MRI images produced by the U-Net model.

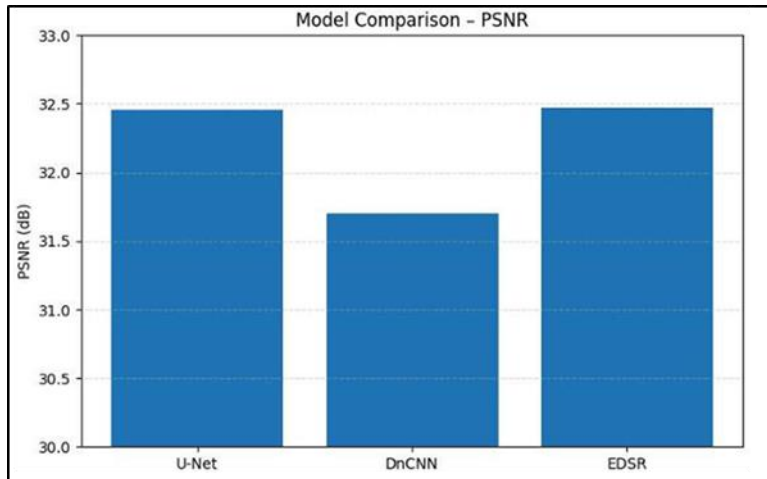


Figure 5.7: PSNR evaluates how well each model lowers noise and improves MRI clarity.

2. Comparison of SSIM

The SSStructural Similarity Index (SSIM) measures the structural detail retention capability of the improved images. Structure includes but is not limited to edges and textures; therefore, from the data shown in this graph we see the following results:

- U-Net (SSIM ≈ 0.9066) does an excellent job of maintaining the tissue boundaries and structural integrity.
- DnCNN (SSIM = 0.8919) achieves this because of its more aggressive noise suppression, which results in lower local contrast.
- EDSR produces a similar SSIM (SSIM ≈ 0.8978), indicating that good structure preservation is achieved with super resolution methods as well.

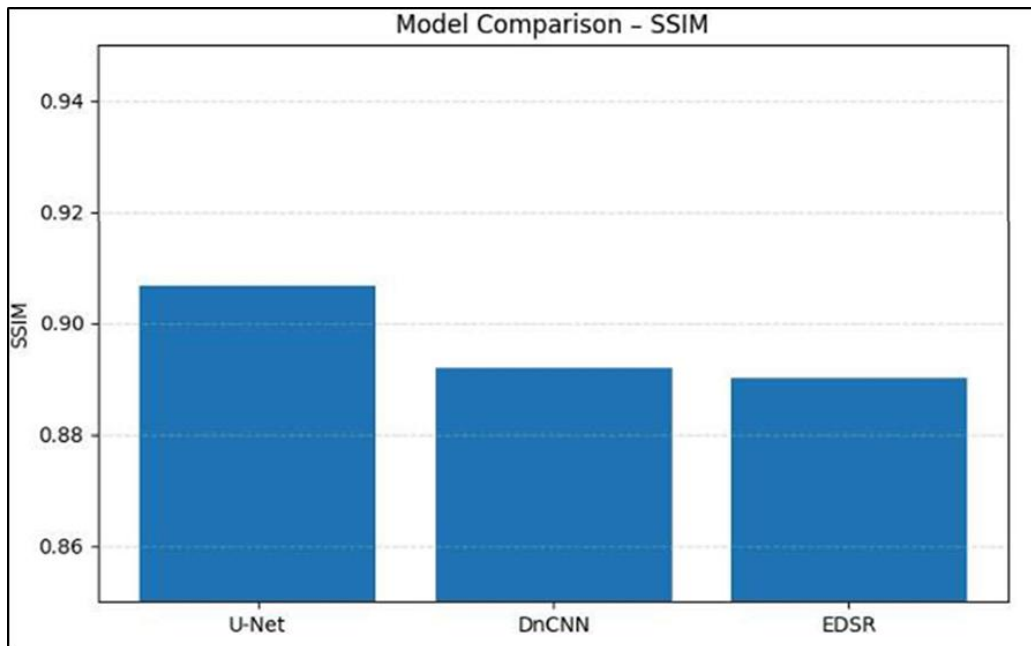


Figure 5.8: SSIM evaluates each model's ability to retain structural information in MRI pictures.

These results demonstrate that EDSR and U-Net have complementary strengths, with EDSR offering the best pixel-level reconstruction quality and U-Net excelling in structure preservation.

5.1.2 Performance of Classification Models

1. Overall Performance of the Model : The five models' accuracy, precision, recall, F1-score, and specificity are contrasted in the bar graph.

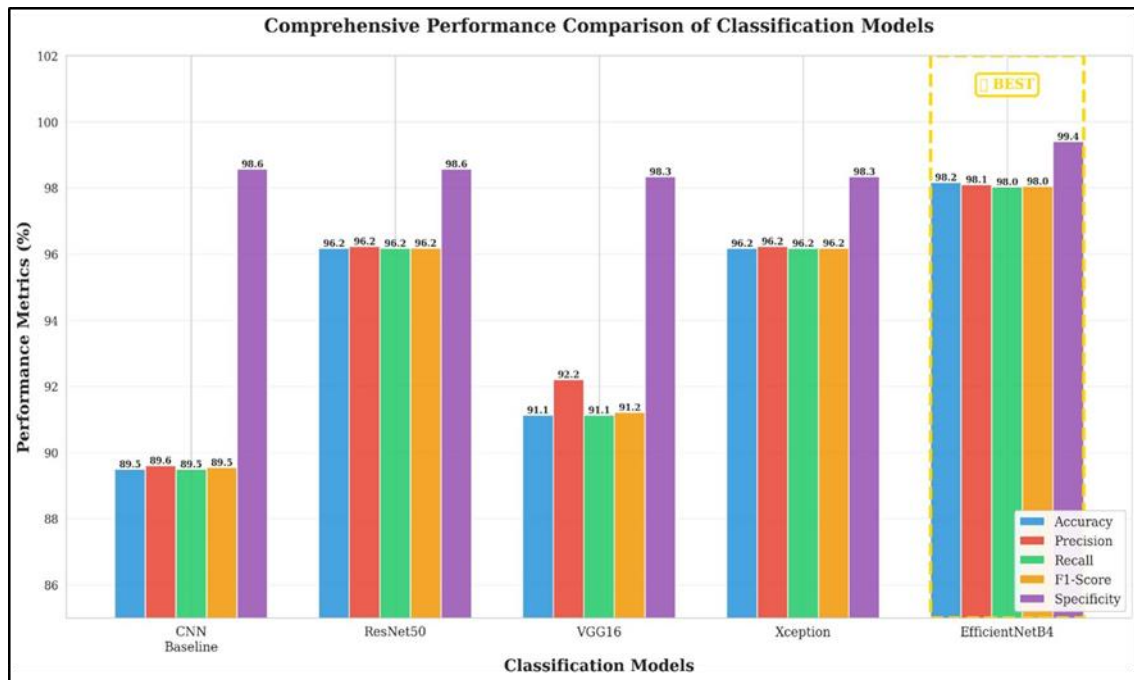


Figure 5.9: Examine the fundamental evaluation metrics for each classification model.

Overall Findings

- An original build of Custom CNN was used to evaluate the impact of transfer learning, resulting in almost 100% improvement over multiple metrics.
- Compared to the other comparative architectures, EfficientNetB4 produced superior performance across all metrics (accuracy, losses and ROC-AUC).
- Trade offs exist between the costs of running a neural network and performance of a neural network on detecting brain tumours - hence why both Xception and VGG16 performed at such high levels and still produced acceptable levels of sensitivity and cost relative to the size of the networks.
- For models like ResNet50, larger datasets are needed to reach maximum effectiveness.
- The current study indicates that custom architectures designed with scalable and efficient use of separable convolutions should provide the best potential for generalisable brain tumour classification performance.

2. Performance of AUC-ROC According to the AUC-ROC analysis, all models have a strong capability of classifying data points ($AUC > 0.97$). EfficientNetB4 indicates that the decision boundaries are almost perfect, as shown by the AUC value

of 0.9993 with very little overlap between classes. ResNet50 and Xception also showed good performance for the class-level separation of tumours and the probability calibration of their outcomes, with AUC values of 0.9950. VGG16 showed good performance (AUC of 0.9820), but due to the age of the architecture and the inability to extract fine details of features, it will have limitations in its ability to classify tumour types. The baseline CNN demonstrated a strong level of performance, expressed as an AUC value of 0.9750; however, because of the shallow depth of the model, it is less capable of classifying tumours than deeper learning algorithms.

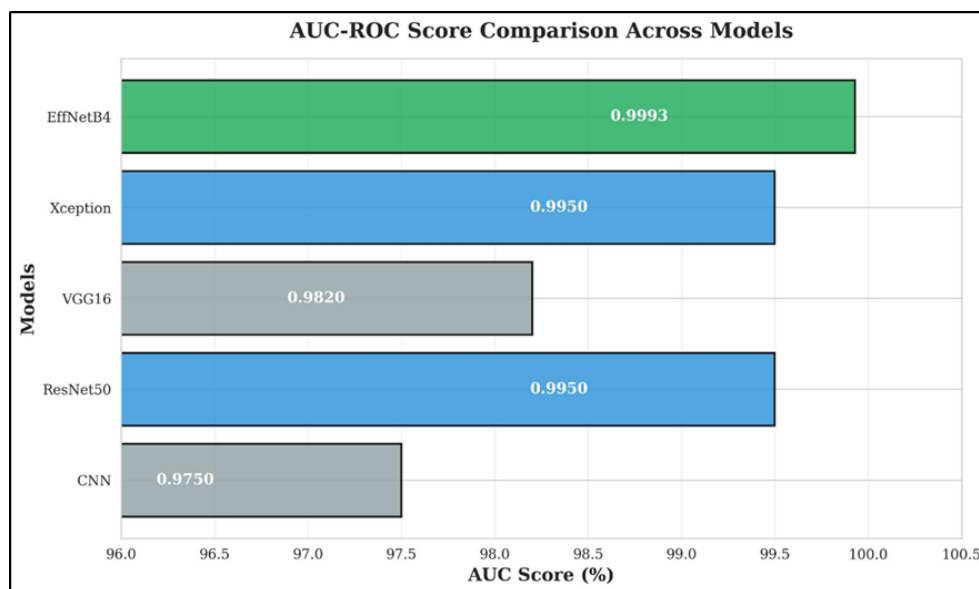


Figure 5.10: Evaluate each model's AUC-ROC scores to determine how well it can differentiate between classes.

All models achieve good results yielding optimal performance across all thresholds, however, EfficientNetB4 is demonstrated as being the most efficient and universally applicable model based on AUC-ROC.

3. Comparison of Radar Plots A radar plot is used to compare the five models regarding their Accuracy, Precision, Recall, F1-Score, and Specificity. EfficientNetB4 performs the best in terms of all metrics and has the largest and most consistent polygon shape. EfficientNetB4 scores consistently high across all metrics, indicating it has performed better

than the other models when comparing overall performance to others using a balanced approach. The second highest scores on the radar plot are from Xception, which has a relatively large radar plot area (~96–97% for most metrics). ResNet50 has also achieved a relatively large radar plot area but is smaller than that of Xception across all areas on the radar plot. VGG16 also performed well, but it was significantly lower than that of the other two models for recall and precision scores. Overall, CNN has received the smallest radar plot area, which indicates that its performance on the five metrics included in this study was generally poor when compared to the other models, except in terms of specificity, in which it performed significantly better than the other four models.

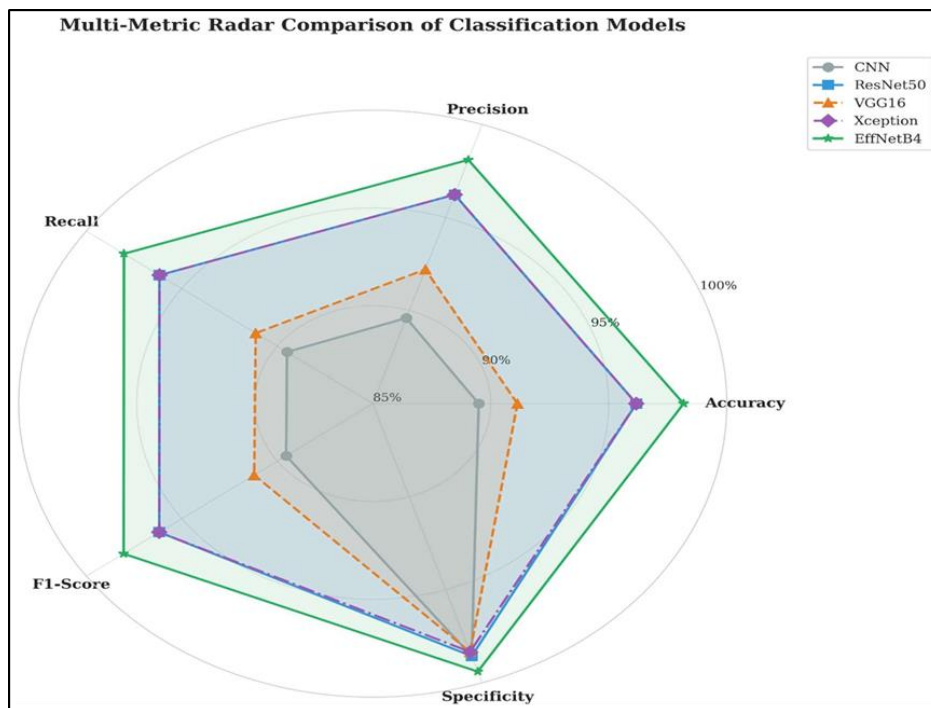


Figure 5.11: Use a single comparative radar plot to see the multi-metric performance distribution of every model.

5.1.3 Testing Web Applications and Verifying Models

The web interface was tested with MRI scans belonging to four groups (pituitary glioma, meningioma, and no tumor) through the EfficientNetB4 model after it was implemented on the website. All scans have a class-based confidence score associated with each one, with the most likely class typically corresponding to the type of tumor found in the experience. The performance of the web-based version of

the model closely mirrors that of the offline version, as evidenced by the dynamically displayed predictions provided through the web interface. Therefore, these tests demonstrate that the deployed web interface can accurately identify multiple different types of tumor patterns in MRI scans and provide consistent and accurate classification accuracy in a production environment.

1. Status of the Web Interface When you first access the web interface, no MRI scans have been uploaded, meaning that the web interface has no content in the classification area. This demonstrates that the application has been designed to wait until a valid input is received to perform any inference on the MRI scan and, therefore, the web application has responded correctly to the event. This also verifies that the model loading process and validation are functioning correctly, as well as that the web interface is functioning correctly.

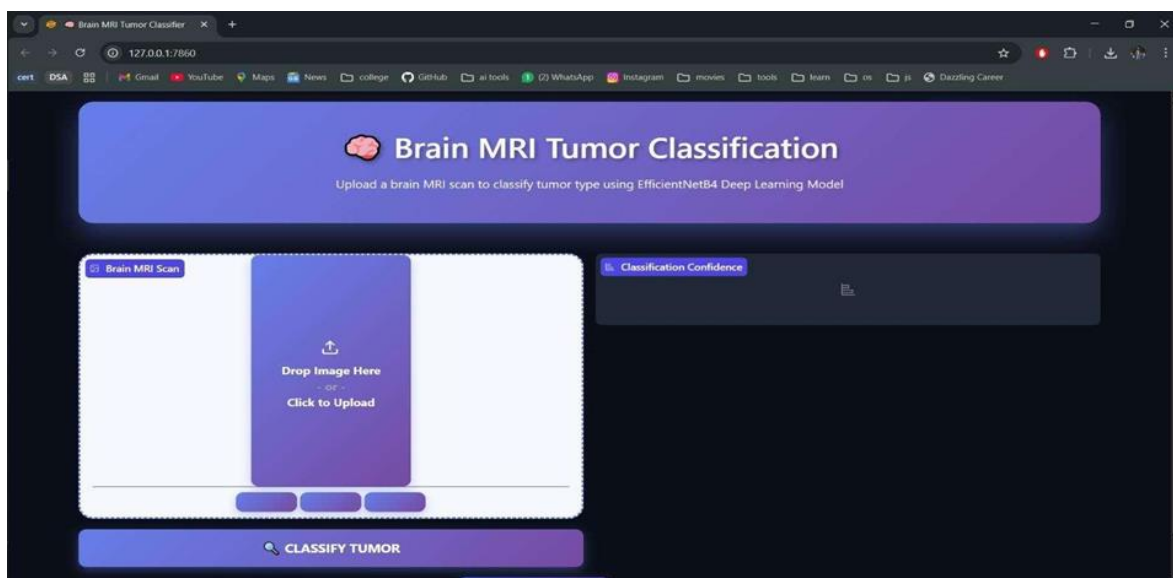


Figure 5.12: The tumour classification system's initial screen while the MRI image is being uploaded

- 2. Prediction of Glioma Tumours** Upon completion of the fourth MRI Scan, it was easy to sight patterns of infiltrative growth associated with a glioma. The computer modelling for gliomas exhibited an 85% confidence level to indicate when a glioma would be produced. Although this confidence level was somewhat lower than those for all other tumour categories, it still closely matched the factual diagnosis. This shows that computer models will provide accurate predictions for gliomas because of their many different appearances, including texture and shape.

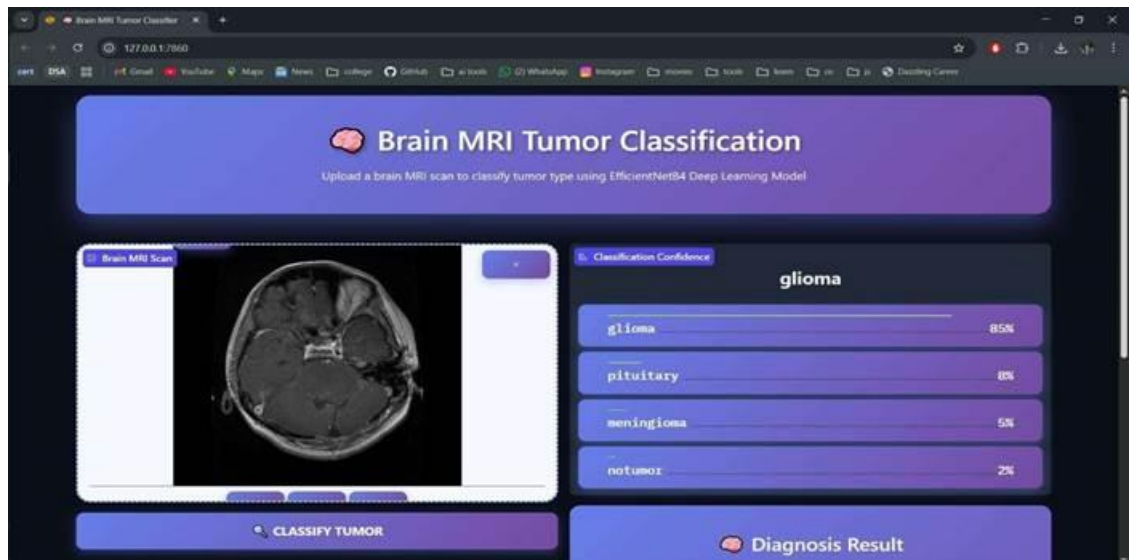


Figure 5.13: A classification interface with matching confidence bars displaying a glioma prediction

3. Prediction of Meningioma Tumours A distinct mass that was indicative of a meningioma was visible in the third image. Meningioma was predicted with 95% confidence by the system, while secondary classes such as glioma and pituitary received only minor confidence values. The model's ability to identify tumor boundaries and dense extra-axial growth patterns characteristic of meningiomas is demonstrated by this successful classification.

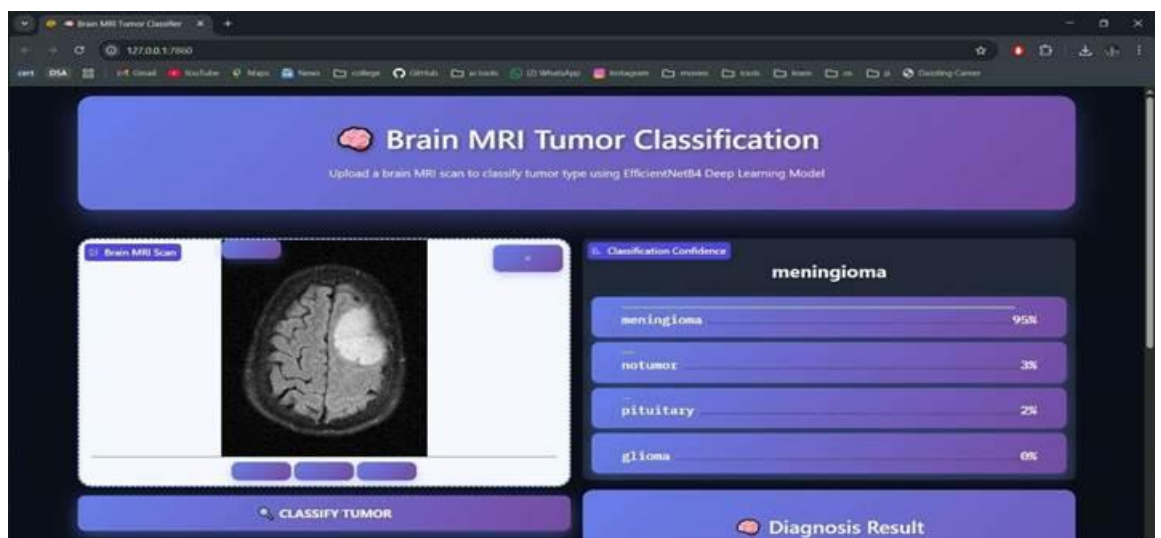


Figure 5.14: A meningioma tumour identification system with a comprehensive confidence distribution.

5.1.4 Medical Assistant web application

1. Landing Page (Before Login)

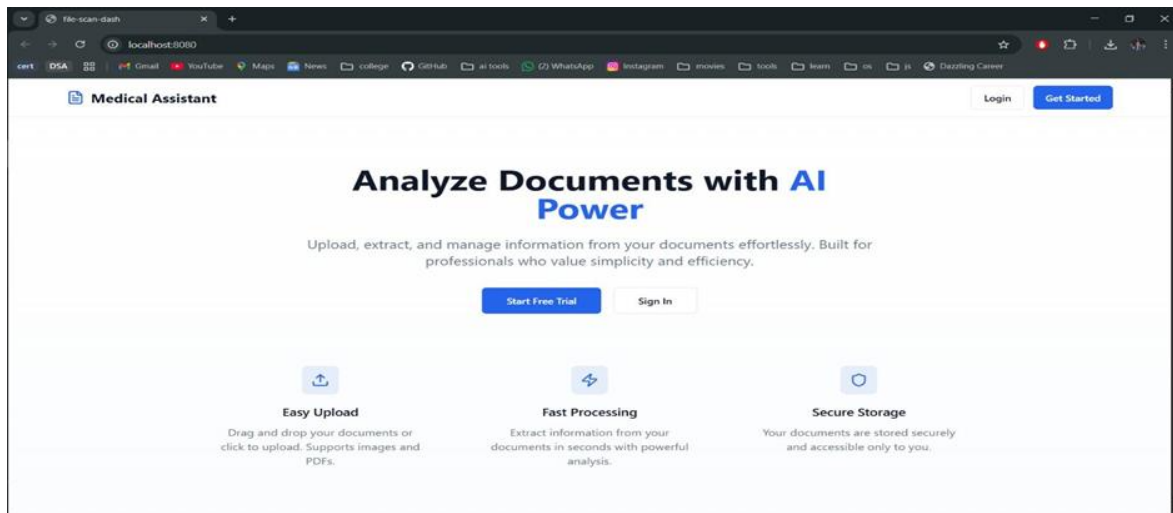


Fig 5.15 : Shows the public entry page where users learn about the platform and start using the AI-powered document analysis service

2. Page of Login The user-friendly, focused, and simple design of the Login page makes logging in easy and confusion-free by providing a straightforward registration process via email and password information, a Sign In Button to authenticate your account, and a "Sign-Up" link for first-time users.

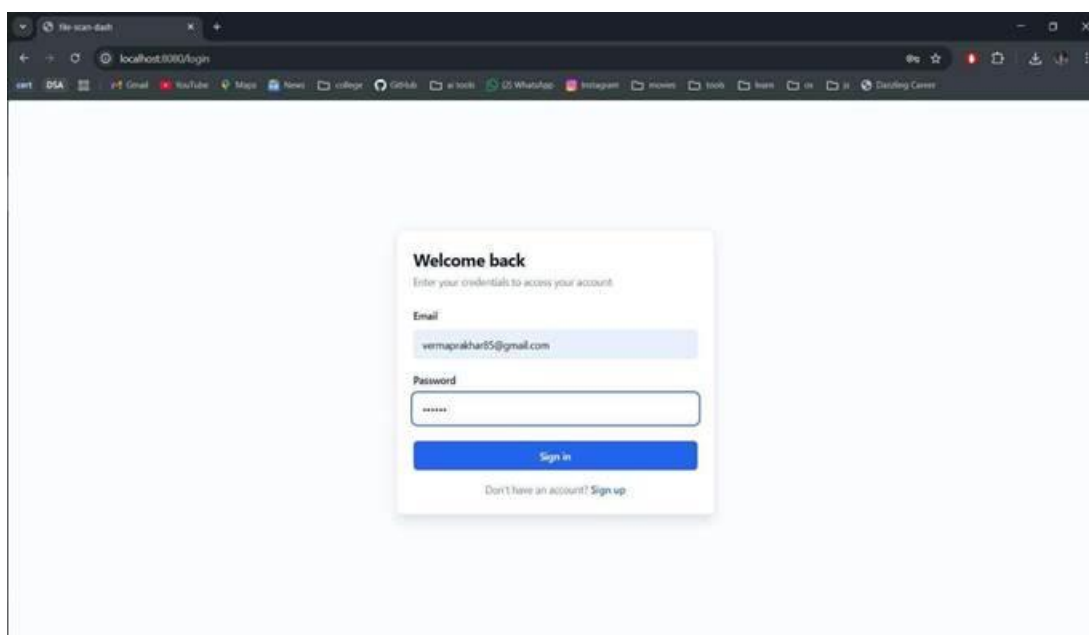


Figure 5.16: Shows the sign-in screen where users can access their accounts by entering their password and email.

3. Dashboard (Login Home Page) When users first log in, they see their dashboard, which serves as a user interface for the system from which they can perform most of their current tasks. In addition, the dashboard provides users with personalized interactive messages when they log into their account, including an initial greeting to the user. In addition to greeting the user, the dashboard shows the user some key information about their activity in the system, including how many documents they have stored in the system and how many documents they have uploaded in the previous month. The dashboard lists the most recent documents that were uploaded, providing a quick reference for users to see what was uploaded most recently. A large quick-upload button makes it easy for users to add a document directly to their account from this page. Overall, the dashboard provides users with easy access to their most important actions and a brief overview of their activities in the system at all times.

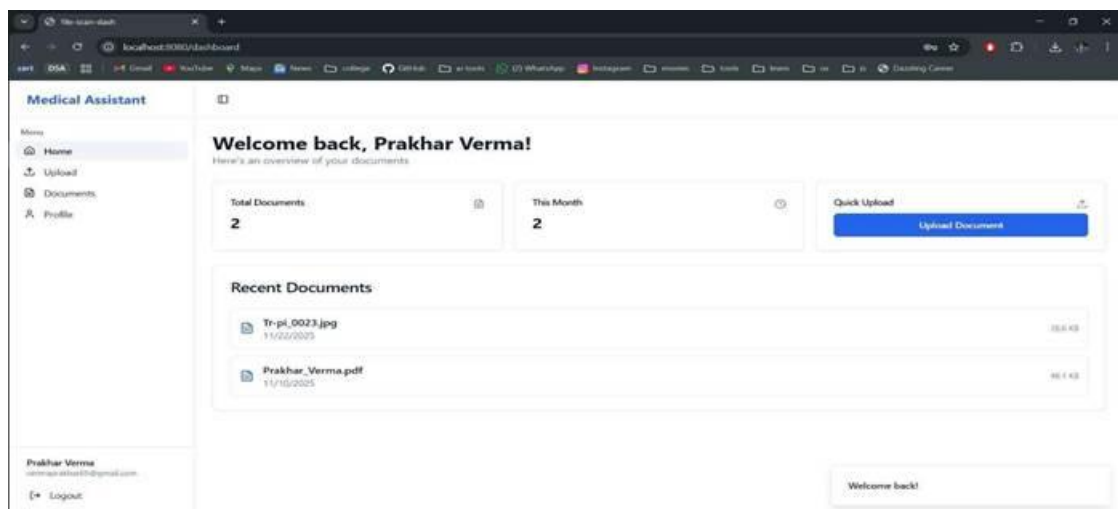


Figure 5.17: Offers statistics, a quick way to upload new documents, and a summary of user activity with recent uploads.

4. Page for Uploading Documents With the Upload Document page, Users can easily upload their own medical documents to their accounts. Users have the option to upload multiple file types including JPG, PNG, WEBP, and PDF files, through a drag-and-drop upload box to put their files either by dragging them to the upload box or by clicking the box and selecting the files manually. Users will see a preview of the file they selected, including its file size and thumbnail image, and also have the option to delete it before submitting the

file for storage on the health care system's server and entering it into an automated AI system to create a report. The goal of this page is to create a user-friendly experience for uploading documents with minimal errors, a visually appealing interface, and a streamlined process for successfully completing the upload of documents.

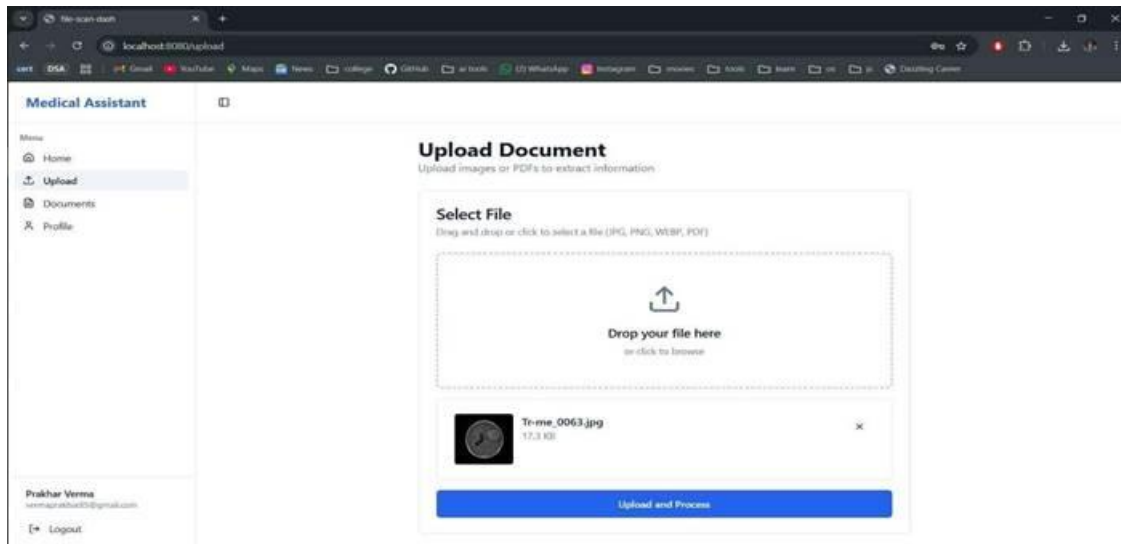


Figure 5.18: Enables users to drag-and-drop medical PDFs or images for AI-based processing

5. Documents Page

The Documents Page is the central repository for all medical records that have been uploaded to the platform. This page allows users to easily search for any uploaded document by entering the document title into the search bar provided. Each document is listed as a "card" that displays the following information about that document: a preview thumbnail image, document title, upload date, file size and buttons for either viewing or deleting that document. After a user has uploaded a document to the platform, a notification will appear confirming successful upload of that document. Overall, the Documents page offers a convenient and streamlined way for users to easily navigate through, view and manage their uploaded documents stored on the platform.

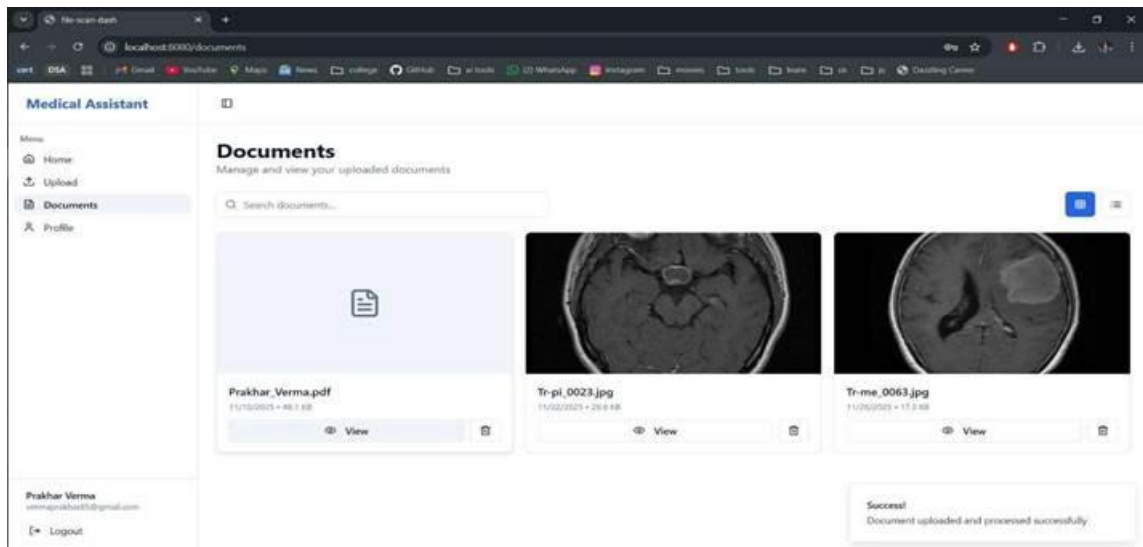


Figure 5.19: Shows all uploaded medical documents with options to search, view, and delete files.

6. Page for Profile Settings

Profile Settings Page displays Storage Usage and Personal Information. Displays verified accounts in an easy to read format. Clearly showing Your email address and name as "read only." In addition, the Page shows your Summary of Document Storage. It lists the total number of documents uploaded and the sum of their Storage Space used; i.e. In this example, three documents, totaling 0.09 MB. An easy-to-use Page allows Users to Manage their Account Information and Monitor the Amount of Storage Space they consume.

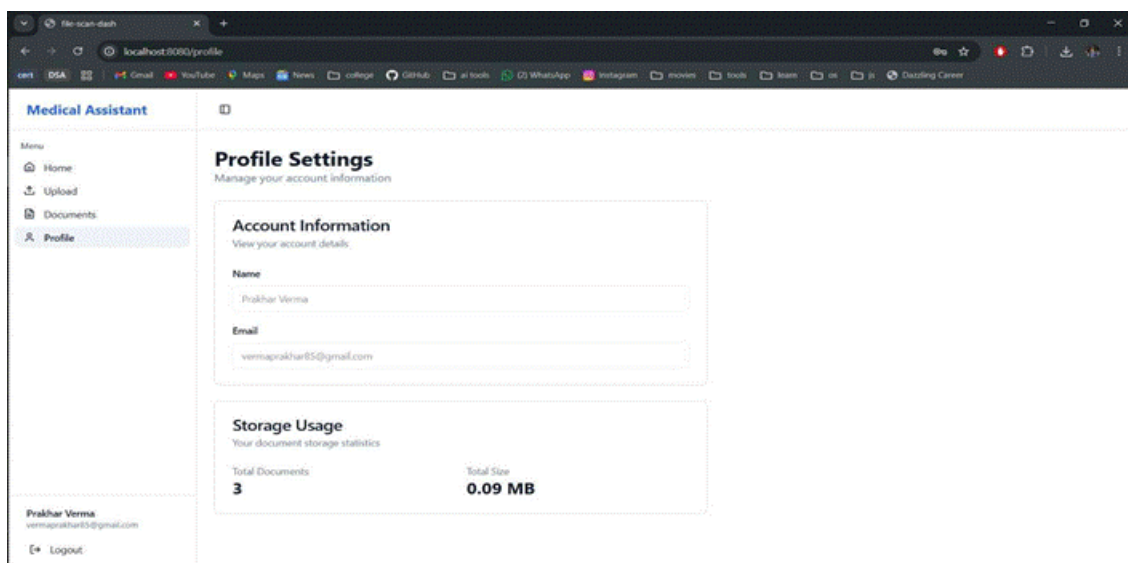


Figure 5.20: A straightforward, well-organised layout that shows the user's personal data and total document storage usage.